Shadi Rahimfallah

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PROFILE

Economist (Ph.D. candidate, York University, expected June 2027). *Research:* firm investment, financial frictions, and technology adoption; electricity market design (PJM capacity market, generator bidding, market power). *Industry and consulting:* built demand, pricing, and cost models, business cases, budgets, and forecasts for CEOs, shareholders, and clients. Stata, Python, SQL. *Areas of interest:* energy, competition, and financial analysis. Authorized to work in Canada.

EDUCATION

Ph.D. in Economics, York University, Toronto — 2022 – June 2027 (expected)

M.Sc. in Energy Economics, University of Tehran, Tehran — 2016 – 2019
Thesis: "Shale Gas and the Export Competitiveness of Iran's Petrochemical Industry"

B.Sc. in Theoretical Economics, Shahid Beheshti University, Tehran — 2012 – 2016

RESEARCH PROJECTS

Growth, Productivity, and Macro-Finance

"Financial Frictions, Used Capital, and Misallocation", *Job Market Paper* [PDF] [SSRN] [Slides] [Code]
Under review, *Journal of Economic Dynamics and Control*

General-equilibrium model with new and used capital and borrowing constraints, solved by dynamic programming; firm fixed-effects regressions on Vietnamese data for calibration: used capital cuts productivity losses by 39%.

➤ Presented at: CEA Annual Meetings (2025, 2026); Macro Development Group, York University (2025, 2026)

"Second-Hand Machines, Firm Capability, and Technology Adoption", draft in progress

Vintage-choice model and firm surveys (15 countries): capability-bound vs. cost-bound users of old machines.

"News, the Economy, and Bond Returns", with Alireza Naeiji, work in progress

Neural-network forecasting: how macroeconomic data and news text predict future Treasury excess returns.

Energy and Electricity Markets

"Electricity Capacity Market", with Jeremy Lin (Johns Hopkins; formerly PJM), work in progress

Large-load market design: how fast-growing data-centre load affects PJM's capacity market, and an auction to serve it.

"Geoeconomics of Global Energy Transformation", with Farkhondeh Jabalameli,
Journal of World Sociopolitical Studies 7(1): 55–87, 2023 [DOI]

ARDL, error-correction, and causality tests: long-run oil pass-through to U.S. polyethylene prices.

Canadian Economy: Public Data Reports

"Canada's Exporters and the U.S. Market, 2000–2025", Statistics Canada data [Report]

85% of goods exporters sell to the U.S. and 65% only there (2025; 56% in 2010); small firms are most dependent.

"How Canadian Firms Work: Evidence from the 2024 Enterprise Survey", survey-weighted [Report]

Obstacles, innovation, finance, trade, and green practices of Canadian firms by size, sector, and region.

PROFESSIONAL EXPERIENCE

Research

Research Assistant, York University — 2023 – 2025

Responsibilities: *Impact evaluation of a World Bank cluster-randomized land-titling trial (Philippines), rebuilt from raw survey data: balance and attrition checks, ANCOVA intent-to-treat effects with clustered standard errors, and tests of consumption insurance; built a World Bank–FAO country–year panel on trade risk and food security.*
Tools & Technologies: *Python, Stata.*

Research Assistant, University of Tehran 2018

Responsibilities: *Studied Iran's day-ahead electricity market design and regulation and analyzed generator bidding data for market power; compared administered and market gas prices; estimated two-way fixed-effects models of how supply shocks and export credit shape Iran's export diversification.*
Tools & Technologies: *MATLAB, Stata, Python.*

Industry and Consulting

Senior Financial & Economic Analyst | Accounting, Partogen Beh Tashkhis 2019 – 2021

Responsibilities: *Led financial and market analysis for the CEOs and shareholders of a four-company diagnostics group; built business cases for new tests, from market sizing and cost per test through pricing; built demand, price, and cost forecasts for budgeting and business planning; led the case for COVID-19 PCR testing on existing equipment, which became a material contributor to revenue.*
Tools & Technologies: *Stata, SQL, Tableau.*

Financial & Economic Analyst | Accounting, Partogen Beh Tashkhis 2016 – 2019

Responsibilities: *Built demand, pricing, and market-sizing models from sales data; tracked competitor prices and market shares; prepared the budgets and financial reports for the laboratory and three affiliated companies.*
Tools & Technologies: *Excel, SQL, Tableau.*

Economic Analyst, Batab Kara, Economic and Investment Consulting Summer 2017

Responsibilities: *Wrote and presented economic and investment reports for client CEOs and strategy teams; built a house-price forecasting model in MATLAB whose scenarios clients used to time investments; assessed an exporting client's trade risk, including exposure to tariffs and export-market concentration.*
Tools & Technologies: *MATLAB, Stata.*

SKILLS

Programming: Stata, Python, MATLAB, Excel (advanced); R, SQL, Git, LaTeX

Economics: General-equilibrium modelling; DCF/NPV and business cases; forecasting

Econometrics: RCTs and ANCOVA; difference-in-differences; panel data; time series; neural networks

Energy & Competition: Capacity markets (Net CONE, VRR); plant dispatch; market power; market shares

Languages: English (fluent), Persian (native)

TEACHING, LEADERSHIP, AND SERVICE

Teaching Assistant, *York University* 2022 – Present
Project Appraisal, Development Policy, Micro and Macro, Mathematics, Labour

Volunteer Instructor, Graduate Quantitative Skills Initiative, *University of Tehran* 2018 – 2020
Collaborated with a team to provide quantitative and computational training to graduate students

Session Chair, *59th CEA Meetings* (2025); Secretary, *York Economics Graduate Student Assoc.* (2024)

GRANTS & AWARDS

LA&PS Dissertation Fieldwork Fellowship, *York University* 2025

MacroFinance Summer School, full scholarship, *Princeton University* 2025

Computational Methods for Macroeconomics, full scholarship, *Econometric Society* 2023

Decision-Making under Risk, full scholarship, *University of Würzburg* 2021

Full Ph.D. Scholarship, *York*; M.Sc. and B.Sc. Full Scholarships (top 1% nationally), *Iran* 2022, 2016, 2012

REFERENCES

Tasso Adamopoulos (Supervisor), *Professor of Economics, York University* aadamo@yorku.ca

Chaoran Chen (Supervisor), *Associate Professor of Economics, York University* chenecon@yorku.ca

Jeremy Lin (Research Collaborator), *Lecturer, Johns Hopkins University; formerly PJM* jlin137@jhu.edu

Yanran Guo (Committee Member), *Assistant Professor of Economics, York University* yrguo@yorku.ca